



4.1.1 Fitting the best line ¶

04.1.1 Launch

fit width

A classic problem is to fit the "best" line through a given set of points: Given

$$\{(\chi_i, \psi_i)\}_{i=0}^{m-1},$$

we wish to fit the line $f(\chi) = \gamma_0 + \gamma_1 \chi$ to these points, meaning that the coefficients γ_0 and γ_1 are to be determined. Now, in the end we want to formulate this as approximately solving $Ax = b$ and for that reason, we change the labels we use: Starting with points

$$\{(\alpha_i, \beta_i)\}_{i=0}^{m-1},$$

we wish to fit the line $f(\alpha) = \chi_0 + \chi_1 \alpha$ through these points so that

$$\begin{array}{rcl} \chi_0 + \chi_1 \alpha_0 & \approx & \beta_0 \\ \chi_0 + \chi_1 \alpha_1 & \approx & \beta_1 \\ \vdots & & \vdots \\ \chi_0 + \chi_1 \alpha_{m-1} & \approx & \beta_{m-1}, \end{array}$$

which we can instead write as

$$Ax \approx b,$$

where

$$A = \begin{pmatrix} 1 & \alpha_0 \\ 1 & \alpha_1 \\ \vdots & \vdots \\ 1 & \alpha_{m-1} \end{pmatrix}, x = \begin{pmatrix} \chi_0 \\ \chi_1 \end{pmatrix}, \text{ and } b = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{m-1} \end{pmatrix}.$$

Homework 4.1.1.1. Use the script in [Assignments/Week04/matlab/LineFittingExercise.m](#) to fit a line to the given data by guessing the coefficients χ_0 and χ_1 .

Ponder This 4.1.1.2. Rewrite the script for [Homework 4.1.1.1](#) to be a bit more engaging...)